Lesson 12.1 - The Chi Square Test

> **Video:** Chi Square Test by The Organic Chemistry Tutor **Chapter:** 12 - The Chi-Square Distribution

The goodness-of-fit test

Does the **observed** distribution of counts across categories match an **expected** distribution?

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

- O = observed count, E = expected count, for each category
- **Degrees of freedom** = $k-1$, where k = number of categories
- It is always a **right-tailed** test: a large χ^2 means the observed counts are far from the expected ones

The chi-square distribution is **not symmetric**; it is skewed to the right. The critical value χ^2_α cuts off area α in the right tail:

- calculated χ^2 > critical value → **rejection region** → reject H_0
- calculated χ^2 ≤ critical value → **do-not-reject region**

Example - which weekday has the most absences?

A principal expects absences to be spread **equally** over the 5-day school week. 100 teachers were asked which day had the most absences. Test at $\alpha = 0.05$.

Day	Mon	Tue	Wed	Thu	Fri	Total
Observed O	23	16	14	19	28	100
Expected E	20	20	20	20	20	100

Hypotheses:

- H_0: absences happen with **equal frequencies** (a uniform distribution)
- H_a: absences do **not** happen with equal frequencies

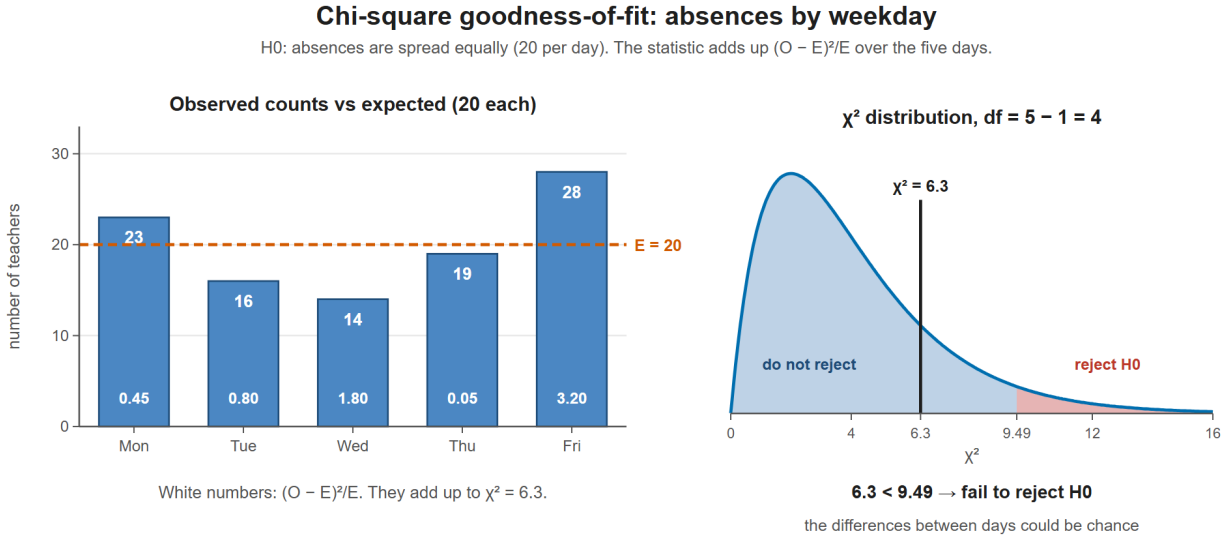
Critical value: df $= 5-1 = 4$. In the χ^2 table with right-tail area 0.05: $\chi^2_{0.05,\,4} = 9.49$.

Test statistic:

Day	$O-E$	$(O-E)^2$	$(O-E)^2/E$
Mon	3	9	0.45
Tue	−4	16	0.80
Wed	−6	36	1.80
Thu	−1	1	0.05
Fri	8	64	3.20
Sum		**126**	**6.30**

$$\chi^2 = \frac{126}{20} = 6.3$$

$6.3 < 9.49$, which is in the do-not-reject region. **Fail to reject H_0.** The variation between days is not large enough to say that absences are unevenly spread across the week.



Left: Friday and Wednesday are furthest from the expected 20, so they contribute the most to χ^2. Right: the total, 6.3, is still below the critical value 9.49, so the uneven pattern could be due to chance.

Key takeaways

- $\chi^2 = \sum(O - E)^2/E$ measures how far the observed counts are from the expected ones.
- Goodness of fit: df = number of categories − 1. It is always right-tailed.
- Large χ^2 (beyond the critical value) → reject H_0 (the data don't fit the expected distribution).

Lesson 12.2 - The Chi Square Distribution Test of a Single Standard Deviation

> **Video:** Chi Square Distribution Test of a Single Variance or Standard Deviation by The Organic Chemistry Tutor **Chapter:** 12 - The Chi-Square Distribution

Test statistic

To test a claim about a population **variance** σ^2 (or standard deviation σ):

$$\chi^2 = \frac{(n-1)\,s^2}{\sigma_0^2}, \qquad \mathrm{df} = n - 1$$

- s^2 = sample variance
- σ_0^2 = the variance claimed in H_0

H_a	Test	Rejection region
$\sigma^2 < \sigma_0^2$	Left-tailed	$\chi^2 <$ critical value, with area α on the left (look up right-tail area $1-\alpha$)
$\sigma^2 > \sigma_0^2$	Right-tailed	$\chi^2 >$ critical value, with right-tail area α
$\sigma^2 \neq \sigma_0^2$	Two-tailed	both tails, $\alpha/2$ each

---

Example - is the spread of exam scores smaller?

The math department's exam scores have $\sigma = 8.6$. A professor thinks it is **less**. A sample of 20 scores has $s = 6.9$. Test at $\alpha = 0.05$.

Known values: $\sigma_0^2 = 8.6^2,\ \ s^2 = 6.9^2,\ \ n = 20,\ \ \mathrm{df} = 19$

Hypotheses:

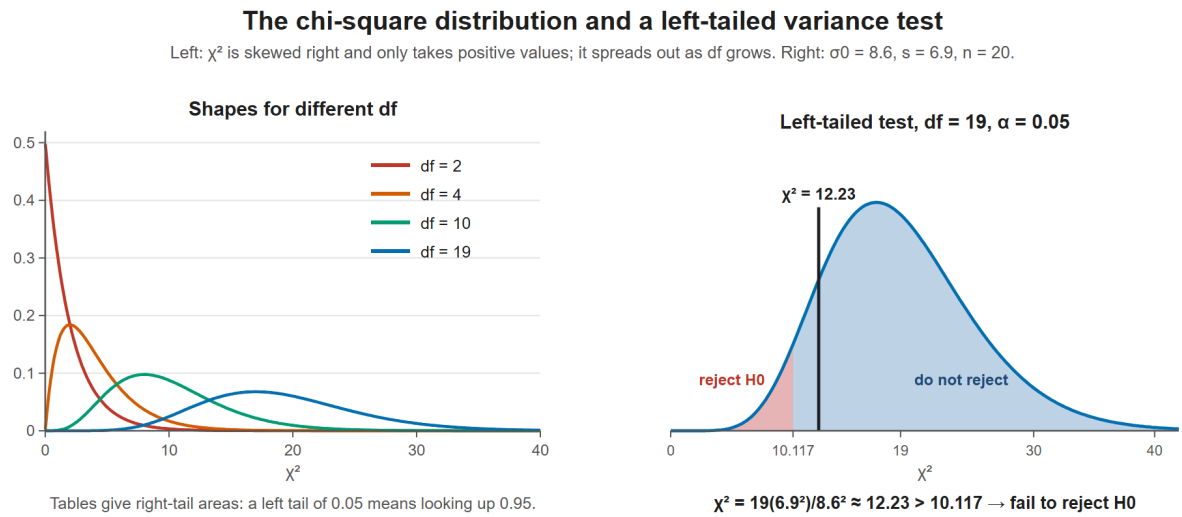
$$H_0 : \sigma^2 = 8.6^2 \qquad H_a : \sigma^2 < 8.6^2 \quad (\text{left-tailed})$$

Critical value: the left-tail area is 0.05, so the right-tail area is 0.95. With df = 19 the table gives $\chi^2 = 10.117$. Reject H_0 only if the calculated χ^2 is **less than 10.117**.

Test statistic:

$$\chi^2 = \frac{19 \times 6.9^2}{8.6^2} = \frac{19 \times 47.61}{73.96} \approx 12.23$$

$12.23 > 10.117$, which is in the do-not-reject region. **Fail to reject H_0.** At the 5% level the sample does not provide enough evidence that the standard deviation is less than 8.6.



Left: χ^2 is never negative and is skewed to the right; with more degrees of freedom it moves right and spreads out. Right: for the left-tailed variance test the rejection region is the left tail below 10.117, and 12.23 does not reach it.

Key takeaways

- Testing a variance or SD: $\chi^2 = (n-1)s^2/\sigma_0^2$ with df $= n - 1$.
- χ^2 tables list **right-tail areas**. For a left-tailed test, look up $1 - \alpha$.
- Left-tailed: reject only when χ^2 is **below** the critical value.

Lesson 12.3 - Tests of Independence Using Chi-Square Distribution

> **Video:** Test of Independence Using Chi-Square Distribution by The Organic Chemistry Tutor **Chapter:** 12 - The Chi-Square Distribution

The idea

A **test of independence** checks whether two categorical variables in a **contingency table** are related.

	Formula
H_0	the variables are **independent**
H_a	the variables are **dependent** (related)
Expected count for each cell	$E = \dfrac{\text{row total} \times \text{column total}}{\text{grand total}}$
Test statistic	$\chi^2 = \sum \dfrac{(O-E)^2}{E}$ over **all** cells
Degrees of freedom	$(r-1)(c-1)$, where r = rows and c = columns

It is a **right-tailed** test: reject H_0 if χ^2 exceeds the critical value.

Example - study time vs type of student

Is the number of hours spent studying each day related to whether a student is a freshman or a senior? $\alpha = 0.05$.

Step 1 - complete the observed table

	0-2 h	2-4 h	4-6 h	**Row total**
Freshmen	76	143	91	**310**
Seniors	147	109	64	**320**
Column total	**223**	**252**	**155**	**630**

Step 2 - hypotheses and critical value

- H_0: study hours are **independent** of student type
- H_a: study hours **depend** on student type
- df $= (2-1)(3-1) = 2$, right-tail area 0.05: $\chi^2_{\text{crit}} = 5.99$

Step 3 - expected counts

Example: $E_{1,1} = \dfrac{310 \times 223}{630} \approx 109.7$

	0-2 h	2-4 h	4-6 h
Freshmen	109.7 (≈110)	124.0	76.3 (≈76)
Seniors	113.3 (≈113)	128.0	78.7 (≈79)

The rounded values still match every row and column total (e.g. 110 + 124 + 76 = 310).

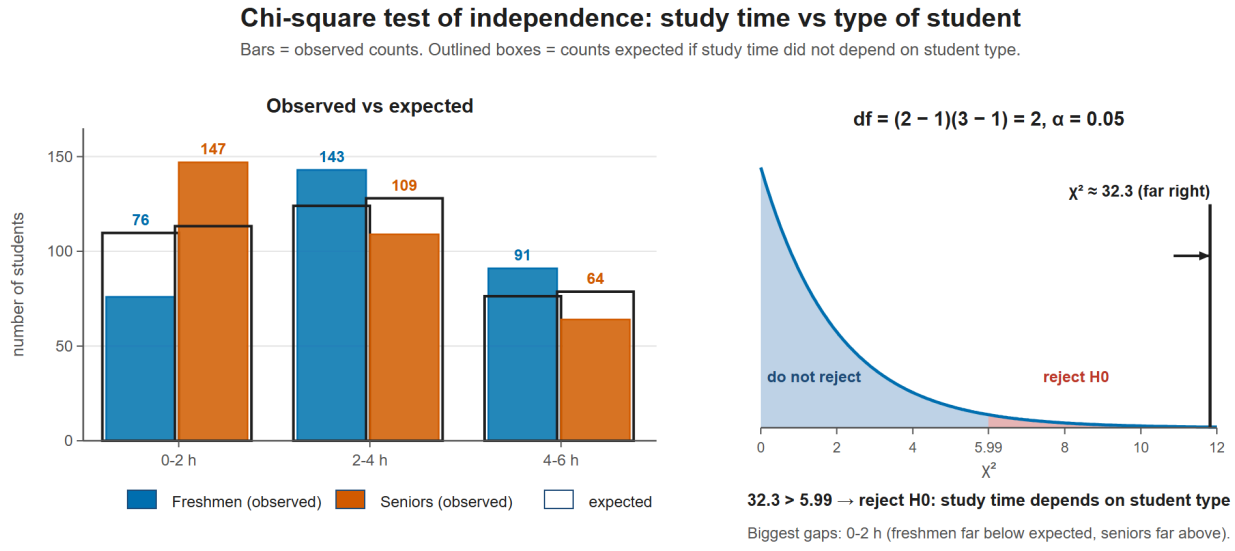
Step 4 - test statistic (with the video's rounded expected counts)

Cell	O	E	$(O-E)^2/E$
Freshmen, 0-2	76	110	10.51
Freshmen, 2-4	143	124	2.91
Freshmen, 4-6	91	76	2.96
Seniors, 0-2	147	113	10.23
Seniors, 2-4	109	128	2.82
Seniors, 4-6	64	79	2.85
Total			**≈ 32.3**

> **Note:** expected counts don't have to be whole numbers. Using the unrounded values (109.73, 124.0, 76.27, 113.27, 128.0, 78.73) gives $\chi^2 \approx 31.7$. The conclusion is the same.

Step 5 - decision

$32.3 > 5.99$, which is in the rejection region. **Reject H_0.** There is strong evidence that study time **depends** on student type. In the data, seniors are much more likely than freshmen to study only 0-2 hours a day.



Left: if study time were independent of student type, the bars would roughly fill the outlined expected boxes. The 0-2 hour group is far off for both freshmen and seniors. Right: $\chi^2 \approx 32.3$ is far beyond 5.99.

Key takeaways

- Expected count = (row total × column total) ÷ grand total.
- $\chi^2 = \sum (O - E)^2 / E$ over every cell, with df $= (r - 1)(c - 1)$.
- Right-tailed: a large χ^2 → reject independence → the variables are related.